

# Let Your Car Listen to Your Respiration Contactlessly with Ubiquitous Acoustic Signals

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## Abstract

Harnessing ubiquitous acoustic signals with capability of contactless respiration detection has the potential to revolutionize lots of intelligent applications. Existing work mainly employs inaudible acoustic signals to detect human respiration. To the best of our knowledge, there is a lack of public datasets due to the uniqueness of such signals. In this paper, we leverage commercial acoustic devices in cars to collect 30.79h respiration data from 23 participants and 30.98h background noise data in various scenarios over six months. An SNR enhancement pipeline is proposed to increase the SNR of data so that the neural network can capture the fine-grained respiration features. Additionally, a data augmentation pipeline is designed to enhance the diversity of the data and improve the generalization ability. Finally, we employ the proposed pipelines to train CNN-based models to detect respiration. Extensive experiments demonstrate our system can achieve a recall rate of up to 97% while maintaining a false positive rate lower than 1.11%.

## CCS Concepts

• **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*.

## Keywords

Contactless Sensing, Acoustic Sensing, Respiration Detection

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## 1 Introduction

Respiration detection is critical in numerous intelligent applications, including active health monitoring [4, 16], sleep tracking [17, 18], and presence detection [7, 19]. Meanwhile, contactless sensing is gaining increasing attention from academia and industry due to its non-intrusive nature, which eliminates the need for individuals to carry any wearable devices, such as smartwatches and pulse oximeters. For instance, ENCAP (European New Car Assessment Programme) will only reward CPD (child presence detection) systems that can detect vital signs including respiration in a contactless manner for car manufacturer from 2025 [1]. Unlike counterparts such as WiFi [9, 13], RFID [5], radar [3], LoRa [20, 21], acoustic-based approach leverages ubiquitous acoustic devices such as speakers and microphones to achieve contactless respiration detection, presenting an appealing alternative with low-cost nature and broad applicability. Existing acoustic-based sensing systems primarily leverage the inaudible acoustic frequency (i.e., above 18KHz) to avoid the effects on human targets [11, 14, 15]. Due to the uniqueness of inaudible acoustic signals, there is a lack of publicly available datasets for contactless respiration detection, particularly in the car environments, seriously hindering the development of data-driven method in the field of acoustic-based contactless sensing.

Specifically, we face the following challenges to take the idea into practice. Firstly, contactless respiration detection depends on the channel variation of acoustic signals induced by chest movement. However, there is no publicly available acoustic-based contactless respiration detection dataset that employs acoustic signals with frequency of above 18KHz. Secondly, the millimeter level chest movement during respiration only introduce very weak channel fluctuation, which makes it difficult to extract effective features when using deep learning methods. Thirdly, it is crucial to enhance the generalization ability of neural network for accurate respiration detection across various practical scenarios.

To address the aforementioned challenges, we employ multiple commodity speaker-microphone pairs equipped within cars to collect acoustic channel data for human respiration in the various real-life scenarios, including underground parking lots in office

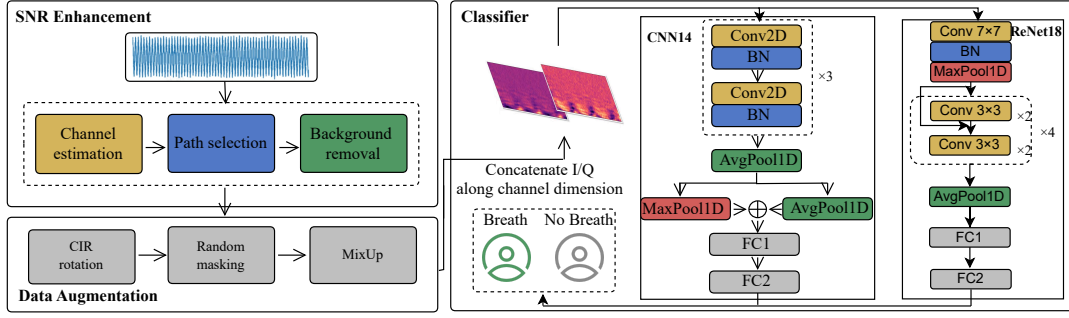


Figure 1: Overview of the proposed respiration detection system.

building and laboratory. 23 participants are recruited to collect 30.79h of respiration data, while 30.98h background data are collected as negative samples in various scenarios. To ensure that the deep neural network can effectively capture the fine-grained respiration feature in channel variation, we propose an SNR (signal to noise ratio) enhancement pipeline to suppress the impact of noise and interference, including path selection and background removal. Path selection is employed to identify the channel paths that contain respiration information and discard the paths that are dominated by noise, while background removal is employed to eliminate the static component in the channel and only retains the respiration component. Additionally, a data augmentation pipeline is designed to improve the generalization ability of the neural network. Considering the nature of complex number for channel signals, a random rotation method is implemented to enhance the diversity of channel data. And the random masking and mixup methods are employed to further augment the diversity of the data. After SNR enhancement and data augmentation, the real and imaginary parts of the channel data are separated into two channels and then concatenated as the input of CNN14 and ResNet18 network to detect respiration. The main contributions of this paper can be summarized as follows:

- We collect 30.79h respiration data from 23 participants and 30.98h background data in various real-life scenarios using multiple commodity acoustic devices equipped within cars. This valuable dataset will be public at an appropriate time.
- We propose an enhancement pipeline to significantly improve the SNR of data, which is proved to be highly effective to extract the fine-grained respiration features. Additionally, a data augmentation pipeline is also introduced to enhance the diversity of the data and improve the generalization ability of the deep neural network.
- Extensive experiments demonstrate we can achieve a recall rate of up to 97% while maintaining a false positive rate lower than 1.11% in real-life environments.

## 2 System Design

In this section, we will first provide an overview of the proposed system. Following this, we will delve into the details of its implementation.

### 2.1 Overview

Fig. 1 illustrates the diagram of the proposed respiration detection system, including three key modules: *SNR Enhancement*, *Data*

*Augmentation*, and *Classifier*. The SNR enhancement module is designed to extract fine-grained respiration features from the received acoustic signals and improve the SNR of channel data. The data augmentation module employs multiple methods to comprehensively increase the diversity of the collected data, enabling the trained model to be easily generalized to complex real-life scenarios. Taking the channel data as input, we fuse the real part and imaginary part of complex number to train the classic CNN14 and ResNet18 neural networks as classifier to accurately identify respiration status.

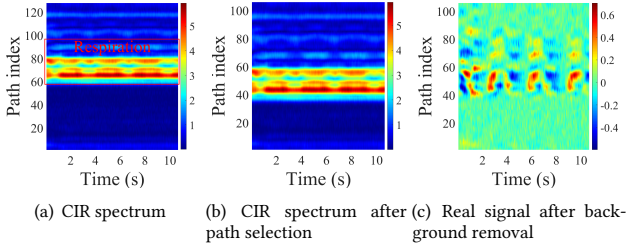
### 2.2 SNR Enhancement

Given that the acoustic-based contactless respiration detection employs the very weak reflected signal induced by the subtle chest movement of human target, it is crucial to enable the neural network to capture the fine-grained features of respiration. To this end, we propose to enhance the SNR of the respiration signal in this section.

**2.2.1 Channel Estimation.** The principle of contactless respiration detection is that the cyclic movement of chest during respiration can periodically change the propagation channel of acoustic signals. To estimate the channel impulse response (CIR), we utilize auto-correlation between the transmitted and received acoustic signals. The CIR characterizes how the acoustic signals propagate through multiple paths in the space. Specifically, we transmit modulated ZC sequences with ideal auto-correlation property from the speaker and receive the acoustic signals with microphone [12]. The center frequency is set to 19kHz, which is inaudible for human targets. Mathematically, the CIR for  $i$ th path at time  $t$  can be denoted as

$$h_i(t) = h_{s_i} + h_{d_i}(t) = A_{s_i}e^{-j\theta_{s_i}} + A_{d_i}(t)e^{-j\theta_{d_i}(t)}, \quad (1)$$

where  $h_{s_i}$  is the static component which can be safely assumed to be constant over a short time period,  $h_{d_i}(t)$  is the dynamic component induced by the movement,  $A_{s_i}$  ( $A_{d_i}(t)$ ) and  $\theta_{s_i}$  ( $\theta_{d_i}(t)$ ) represent the amplitude and phase of the static (dynamic) component, respectively. The CIRs of all paths constitute the channel spectrum that provides physical space information in environments, including the human target in the space. Fig. 2(a) gives an example of channel spectrum when a target breathes normally. We can clearly observe the periodic fluctuations induced by respiration, which is indicated with a red box.



**Figure 2: An illustration of SNR enhancement.**

**2.2.2 Path Selection.** Guided by the feature engineering that proves the effective features can greatly improve the performance of deep learning, we firstly aim to eliminate the disturbance of non-relevant features and enhance the SNR for contactless respiration detection. Fig. 2(a) indicates not all paths in channel spectrum contain respiration information. The paths without respiration will interfere with the respiration feature extraction for the neural network. Based on the observation that CIR will experience greater attenuation for paths with longer length, we take the path with maximum amplitude as the first path and discard the paths that exceed a certain threshold length of 12m. The channel spectrum after path selection is shown in Fig. 2(b). With the effective path selection, the SNR is enhanced to enable reliable contactless respiration detection.

**2.2.3 Background Removal.** In addition to path selection, we also fully exploit the sensing capability of CIR spectrum by enhancing signal features of respiration. Since only the dynamic components in the CIR demonstrate the human target's respiration, the static components can be considered as interference that hinders the extraction of respiration related features. Thus, to further enhance the SNR of channel spectrum, we employ a loopback filter [2] to eliminate the influence of static component in the CIR and concentrate on the dynamic component that carries respiration information. Specifically, the static component for  $i$ th path at time  $t$  can be estimated as

$$h_{s_i}(t) = \beta h_{s_{i-1}}(t) + (1 - \beta)h_i(t), \quad (2)$$

where  $h_i(t)$  is the estimated CIR for  $i$ th path at time  $t$ , and  $\beta$  is the weight factor. We set  $\beta$  empirically to 0.9. Then the dynamic component for  $i$ th path at time  $t$  can be obtained as

$$h_{d_i}(t) = h_i(t) - h_{s_i}(t). \quad (3)$$

Fig. 2(c) depicts the dynamic component of channel spectrum, where the respiration features can be clearly observed.

## 2.3 Data Augmentation

**2.3.1 Random Rotation.** In our case, there is no publicly available dataset and data collection is highly labor-intensive. We propose to perform data augmentation to increase the diversity of our dataset. Given that the CIR involves complex numbers, we can introduce additional phase shift to rotate the dynamic component. For instance, by introducing a phase shift of  $\theta_r$ , we can rotate the dynamic component to

$$h_{d_i}^{\wedge}(t) = h_{d_i}(t) \cdot e^{j\theta_r}. \quad (4)$$

In order to balance computation consumption and data diversity, we adopt a strategy of randomly rotating each data sample twice, with a phase shift between  $0 \sim 2\pi$ . By doing so, we effectively triple the size of our dataset.

**2.3.2 Random Masking.** To prevent overfitting and enhance the generalizability of the model, we employ a random masking technique to mask the channel features in both path index and time domain. We randomly select a starting index  $i_o$  in the range of  $[1, I - \Delta i]$  with equal probability, where  $I$  is the total number of paths after path selection, and  $\Delta i$  is the width of masking. Then we mask a continuous set of paths ranging from  $i_o$  to  $i_o + \Delta i$ . Similarly, in time domain, a starting time point  $t_o$  is selected from the range of  $[0, \tau - \Delta t]$  with equal probability, where  $\tau$  is the total number of time samples, and  $\Delta t$  is the time width of masking. We mask a consecutive time sequence ranging from  $t_o$  to  $t_o + \Delta t$ .

**2.3.3 Mixup.** Mixup is a data augmentation technique that involves creating new training samples by linearly interpolating between existing sample pairs and their corresponding labels. Given two examples  $(x_i, y_i)$  and  $(x_j, y_j)$  from the training set, a new sample  $(\tilde{x}, \tilde{y})$  is created as follows:

$$\begin{cases} \tilde{x} = \lambda x_i + (1 - \lambda)x_j, \\ \tilde{y} = \lambda y_i + (1 - \lambda)y_j, \end{cases} \quad (5)$$

where  $\lambda$  is a mixing coefficient between 0 and 1 that follows a certain probability distribution. By randomly generating these coefficients and creating new examples in this way, mixup encourages the model to learn more robust and generalizable representations of the data. In our implementation, we utilize a Beta distribution  $\text{Beta}(\alpha, \beta)$  for  $\lambda$ , where both  $\alpha$  and  $\beta$  are set to 1 in our experiment. In addition, we also include the original data samples in the training set to increase the number of data samples.

## 2.4 Classifier

To verify the effectiveness of the proposed pipeline, we select two classic CNN-based neural networks, i.e., CNN14 [8] and ResNet18 [6], to detect respiration status. We chose these two models based on their widespread adoption and their capacity to extract spatial features. In CNN14, the basic module that consists of a Conv2D layer with a  $3 \times 3$  kernel size and a Batch Normalization (BN) layer is repeated 6 times. Subsequently, a global pooling strategy is designed to condense the feature maps into a fixed length vector. The first Fully Connected (FC) layer further extracts deep embeddings, while FC2 outputs the detection results. ResNet18 is composed of an 17-layer deep convolutional neural network and 1-layer fully connected layer. The kernel size is  $7 \times 7$  for the first convolutional layer and  $3 \times 3$  for the following 16 convolutional layers. Then average pooling is performed and two FC layers are employed to output the respiration results. Note that we add an extra FC layer to accommodate the respiration detection task.

In our implementation, the length for each frame of acoustic signal is 2048, thus the time dimension of channel spectrum for 10s data with 48kHz sampling rate is approximately 243. After the aforementioned processing pipelines, we concatenate the real part (I signal) and the imaginary part (Q signal) as model input with

dimension of  $234 \times 106 \times 2$ . The data is then feed into CNN14 and ResNet18 models for training.

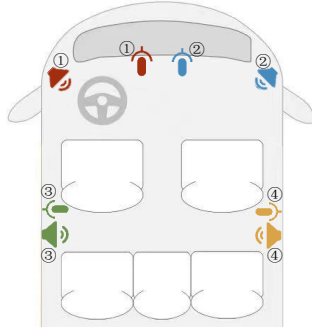
### 3 Dataset

Table 1 summarizes the information of our dataset that totally consists of 30.79h respiration data and 30.98h background data. The duration of each data sample is 10s. To build our dataset, we recruit 23 adults (10 females and 13 males) aging from 20 to 40 to collect respiration data. Each participant is instructed to breathe normally during the data collection process. We also collect background noise as negative samples that there is no respiration. The data collection lasts six months in various practical scenarios including busy parking lots and laboratory and involves multiple commodity acoustic devices (i.e., 12 speakers and 12 microphones) that are installed in 3 cars. For each data collection environment, we select one placement of speaker-microphone pair from the four candidates illustrated in Fig. 3. To ensure that the trained model can work well for complex practical scenarios, we divided the dataset by collection time, such that the training set and test set do not contain data from the same day. Additionally, the test set comprises participants who were not included in the training set. This data division strategy guarantees that the trained model can generalize well beyond the training data.

**Table 1: Dataset information**

|       | Resp   | BG     |
|-------|--------|--------|
| Train | 11.24h | 21.69h |
| Test  | 19.55h | 9.29h  |
| Total | 30.79h | 30.98h |

“Resp” and “BG” are abbreviations for “respiration” and “background”, respectively.



**Figure 3: Experimental setup.**

## 4 Evaluation

### 4.1 Experimental Setup

We implement our system based on Python 3.10 and build the neural network components on PyTorch 2.3.1 [10]. The learning rate is set to  $1e-4$ , while the batch size is 32. Model training is conducted on a computer equipped with an Intel i7-14700K CPU, 64GB RAM, and a GeForce RTX 4070 graphics card. We trained all models for 30 epochs. And we employ recall rate and false positive rate to evaluate the performance of your system. Recall rate is defined as the ratio of true positives (TP) to the sum of true positives and false negatives (FN), while false positive rate (FPR) is defined as the ratio of the number of false positives (FP) to the sum of false positives and true negatives (TN).

### 4.2 Overall Performance

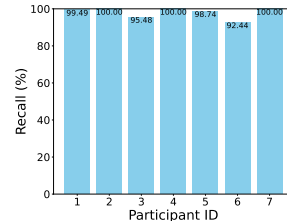
The overall performance is illustrated in Table 2. The experimental results demonstrate CNN14 achieves a remarkable recall rate of up to 97% and an exceptionally low false positive rate of less than 1.11%, whereas ResNet18 attains a recall rate of up to 94.83% and a false

positive rate of less than 5.20%. The slight performance degradation is due to the fact that ResNet18 is more suitable for vision tasks. We also conduct ablation experiments to verify the effectiveness of the proposed SNR enhancement and data augmentation pipelines. Compared with the baseline that directly take CIR as input, the background removal module can significantly improve the recall rate and reduce the FPR. Notably, while path selection marginally reduces Recall, it effectively lowers the FPR. In summary, the results indicate effective preprocessing is essential to extract informative features, which in turn substantially improve the performance of neural networks.

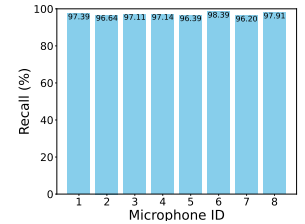
**Table 2: Ablation study results(%)**

| Module Selection     | CNN14             |                  | ResNet18          |                  |
|----------------------|-------------------|------------------|-------------------|------------------|
|                      | Recall $\uparrow$ | FPR $\downarrow$ | Recall $\uparrow$ | FPR $\downarrow$ |
| CIR (Baseline)       | 83.82             | 82.87            | 91.34             | 98.65            |
| + Path selection     | 79.56             | 68.44            | 97.55             | 99.79            |
| + Background removal | 98.05             | 3.74             | 96.34             | 9.20             |
| + Masking and Mixup  | 97.77             | 2.72             | 92.70             | 5.74             |
| + Channel rotation   | <b>97.15</b>      | <b>1.11</b>      | <b>94.83</b>      | <b>5.20</b>      |

Fig. 4 depicts the CNN14 model’s performance on different participants in the test set. We note that the trained model can achieve a recall rate of above 92% for all the participants, showcasing our system can work well for different users. The lowest recall rate happens for a thin female with a small reflection surface. We also evaluate the model’s performance on different microphones in the test set, shown in Fig. 5. The recall rate is above 96% for all microphones, demonstrating our system is robust for the device diversity.



**Figure 4: The impact of participants.**



**Figure 5: The impact of microphone.**

## 5 Conclusion

This paper presents a novel contactless respiration detection system that leverages a deep neural network to detect respiration from channel information extracted from acoustic signals in the car. A valuable dataset is constructed from various real-life scenarios. We address the low SNR challenge by introducing a preprocessing pipeline. In addition, data augmentation pipeline is proposed to enhance the diversity of data and thus improve the generalization ability. Finally, both CNN14 and ResNet18 neural networks are trained to identify respiration using the collected dataset. The experimental results indicate that the proposed pipelines can significantly improve the performance of neural networks, showcasing the potential of using ubiquitous acoustic devices for timely and accurate respiration detection in the car environments.

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